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Electric Power Systems Research 40 (1997) 75–83

ELECTRIC POWER SYSTEMS RESEARCH

Distribution system reliability evaluation using the Monte Carlo simulation method

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Received 21 June 1996; accepted 3 July 1996

Abstract

With the advent of high-speed computers, the utilization of Monte Carlo simulation techniques in reliability evaluation has increased considerably during the last few decades. This paper describes a Monte Carlo approach to a comprehensive distribution system reliability evaluation. The paper presents the results of load point reliability indices and system performance indices for six cases using the Monte Carlo method for a distribution test system. In particular, the probability distributions of system performance indices such as SAIFI, SAIDI and CAIDI for the six cases are presented. The probabilities of these indices occurring during different intervals are analyzed and the effects of different operation philosophies on the system performance indices are illustrated from the probability distribution point of view. The results obtained by the simulation method are compared with those obtained using an analytical approach. This paper illustrates that Monte Carlo simulation is a useful and powerful tool for comprehensive distribution system reliability assessment. The study results show that the simulation method gives acceptable reliability indices and can also be used to provide information on the probability distributions associated with the predicted indices. © 1997 Elsevier Science S.A.

Keywords: Distribution reliability; Probability distributions; Customer service reliability

1. Introduction

Reliability evaluation of individual customer service continuity levels is an important aspect which can significantly affect the design and operating characteristics of power systems in general, and distribution circuits in particular. Analysis of customer failure statistics [1] indicates that distribution systems make a significant individual contribution to overall customer supply unavailability. Approximately 80% of all customer interruptions occur due to failures in the distribution system [1]. Statistics such as these reinforce the need to conduct quantitative adequacy assessments not only for average indices but also for their distributions in the distribution functional zone.

A basic distribution circuit normally involves primary or main feeders, lateral distributors, distribution transformers, disconnects and fuses. The individual load points are connected to the main feeder by the lateral distributors with low-tension distribution transformers at their ends. The outage duration and the number of customers affected due to component failures are reduced by using protection and sectionalizing schemes to isolate the faulted section and restore supply to the healthy sections. The time taken by such isolation and switching actions is generally termed switching/sectionalizing time. It is assumed in all the studies reported in this paper that these actions are manual and not automatic. Provision of an alternative supply is made in some systems in order to restore supply to certain load points while the faulted section of the main feeder is being repaired. Fuses are also usually provided on the lateral distributors to clear local faults. The primary customer load point adequacy (reliability) indices are the average failure rate λ, the average outage duration r and the annual outage duration U [1,2]. The most common system performance indices are the system average interruption duration index (SAIDI), the system average interruption frequency index (SAIFI), the customer average interruption duration index (CAIDI), the average service availability index (ASAI),

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PII S0378-7796(96)01104-2

the average service unavailability index (ASUI), the energy not supplied (ENS), the average energy not supplied (AENS), etc. [1,2].

Conventional reliability analyses are normally only concerned with evaluating the expected or average value of a particular index. Despite the considerable growth in the field of distribution system reliability assessment, service continuity indices [2] continue to be measured almost exclusively by average indices. The actual shape of the statistical distributions associated with these indices is generally not considered. For example, when the frequency of failures at a load point is determined, typically only the average value is calculated and the probability of the load point failing a specified number of times in a given period is normally not evaluated. Due to the random nature of the indices it is very important to evaluate the probability distributions associated with the indices of interest because considerable additional information regarding the variation of the indices around their mean values can be obtained from these probability distributions. These distributions can provide such information as the number of interruptions greater than specified values, the number of customers not restored within certain time periods, etc. They can also be used to more accurately estimate user interruption costs and to indicate the error inherent in using average indices only [3]. Increased awareness of the need for this type of information has, however, resulted in some developmental work in this area over the last decade [1,3–5]. Monte Carlo simulation (MCS) can therefore be applied to a wide range of problems. This paper illustrates the evaluation of individual load point and system performance reliability indices not just average values but also their distributions using MCS. The results obtained using the Monte Carlo approach to analyze RBTS-Bus 4 distribution systems [6] are compared with those obtained by an event based analytical approach [2,6,7].

2. Statistical evaluation of distribution system reliability indices

The MCS method can be used to overcome the difficulties that have been found in developing models to evaluate the probability distributions using the analytical approach [4]. This approach is based on random (nonsequential) simulation. All the studies reported in this paper assume that the component times-to-failure are exponentially distributed because it is assumed that the components operate in their useful or operating life period. The outage times are also assumed to be described by exponential distributions in order to compare with the results obtained using the analytical method.

In this approach, an artificial history of the system reliability is generated using random number generators and the probability distributions associated with the system elements. The simulation process consists of the following steps.

(a) Consider the simulated operational life in years. In this paper, the number of years for each simulation has been assumed to be 1 year.

(b) Assume that both the up time and the down time of each component are independent of each other.

(c) Identify the minimal cut-set for each load point being assessed. Generate different random numbers for both the up and down times of each component belonging to the minimal cut-set.

(d) Convert [1,8] these uniform random numbers into values for time to failure and time to repair using the appropriate distribution [1]. For each component belonging to a minimal cut-set, verify if their times to failure are less than the interval considered in step (a). If this is the case, these 'up times' must be considered and the 'down times' evaluated.

(e) Count the up times and also the down times of such minimal cut-sets in order to find the up times and the down times for the load point being assessed.

(f) Repeat all these steps for a number of simulations until the estimated variables are considered acceptable or an appropriate stopping rule is satisfied. Evaluate the load point indices of interest for each load point as follows:

$$U_T = \frac{\sum r_{ij}}{NS} \text{ h year}^{-1}, \quad i = 1,\ldots, ND,\ j = 1,\ldots, NS \tag{1}$$

$$\lambda_T = \frac{\sum \lambda_{ij}}{NS} \text{ occ. year}^{-1}, \quad i = 1,\ldots, ND,\ j = 1,\ldots, NS \tag{2}$$

$$r_T = \frac{U_T}{\lambda_T} \text{ h occurrence}^{-1} \tag{3}$$

where NS is the total number of simulations, ND is the total number of down states in each simulation for the load point assessed, r_{ij} is the outage time per simulation and per load point, λ_{ij} is the failure rate per simulation and per load point, U_T is the total load point annual outage time, λ_T is the total load point failure rate, r_T is the total load point outage time.

(g) Each simulation produces an estimate for each of the load point reliability indices. The estimated values are used to evaluate the system performance indices (SAIFI, CAIDI, etc.) in each simulation according to the equations presented in [1,2].

(h) The values of the indices in each simulation are arranged in groups within some suitable intervals in which the indices vary. A frequency histogram is created using the number of occurrences of the indices in

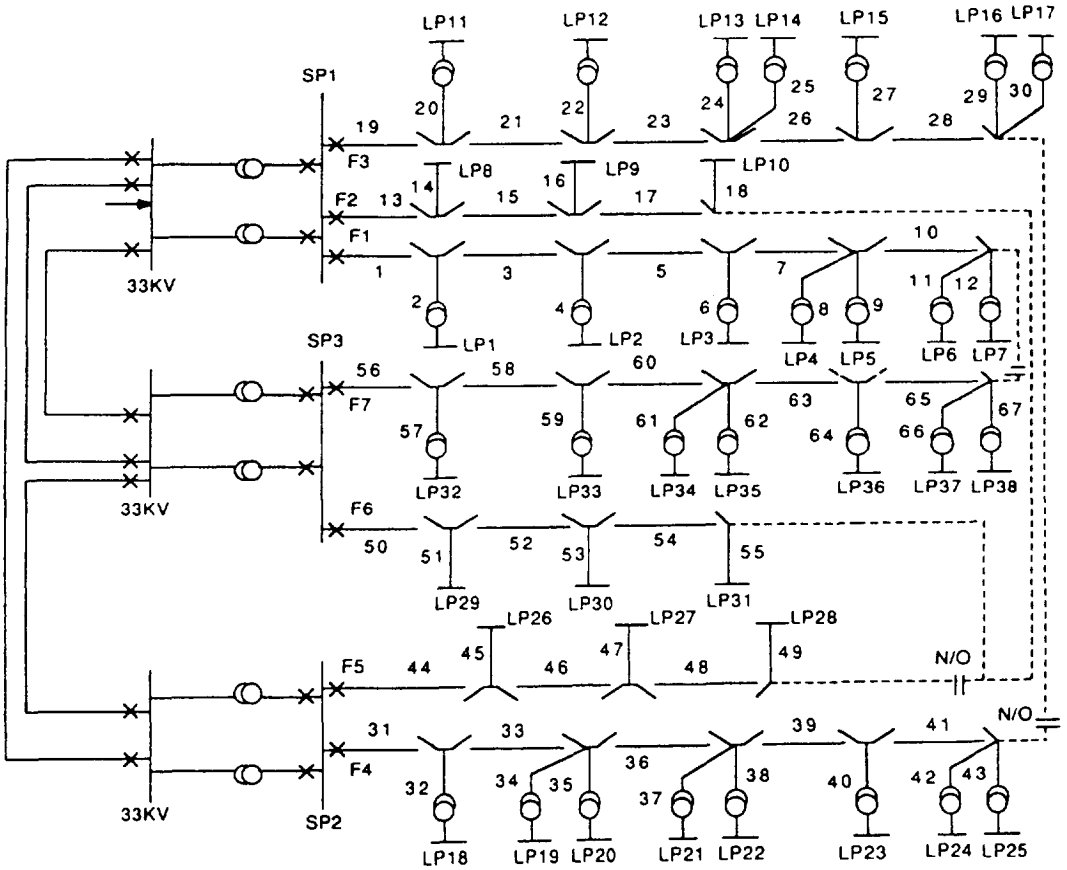


Fig. 1. Single line diagram of the test system.

each class as the ordinate values. The relative histogram is obtained by dividing the class frequencies by the total number of occurrences. These frequency histograms provide valuable information pertaining to the random behavior of the system.

3. Simulation studies

The application of the above approach is illustrated in case studies for the distribution system RBTS-Bus 4 [6]. The single line diagram of this test system is shown in Fig. 1. The peak loading level of bus 4 is 40 MW which is deemed to justify the presence of a 33 kV ring linking three supply points, SP1, SP2 and SP3. The fusegear and disconnects in the distribution system are assumed to be 100% reliable and the alternate supply is assumed to be 100% available, if present. The main and lateral sections of the distribution system are assumed to be composed of either overhead lines or underground cables. The customer data, feeder data and component reliability data for this system are given in [6].

The two sets of reliability indices for the distribution network were evaluated, using Eqs. (1)–(3), for the distribution system assuming that the 11 kV source breakers operate successfully when required to do so. Six case studies were conducted on the 11 kV feeders recognizing disconnects in the main feeders, fuses in the laterals and alternative back-up supply. The effects due to replacing the low voltage transformers with spares as opposed to repairing them were also investigated.

The basic load point indices for all load points for the base case (Case A), i.e., feeders with disconnects, fuses, alternative supply and low voltage transformer repair, using MCS, are shown in Table 1. The system performance indices and each supply feeder performance indices for all six cases are shown in Table 2 (the description of the 6 cases is also provided in this table). Tables 1 and 2 compare the indices obtained by MCS and the analytical method for the RBTS-Bus 4. From these tables, it can be seen that the MCS method provides results which are quite comparable with those from an analytical method. The maximum difference (9.37%) in load point reliability indices occurs at load point 30 for the line system (for the annual unavailability). Their actual values, however, are very close, 0.4 h and 0.44 h for analytical and MCS values respectively. All the percentage differences in these tables are with respect to the analytical values. The system perfor-

Table 1
Load point indices for the cable and line systems

BUS 4	CABLES (MCS)			DIFFERENCE			LINES (MCS)			DIFFERENCE		
Load Point	λ	r	U	λ	r	U	λ	r	U	λ	r	U
	(f/yr)	(h)	(h/yr)				(f/yr)	(h)	(h/yr)			
Feeder 1												
1	0.188	22.56	4.24	0.45%	1.31%	1.88%	0.297	11.82	3.51	0.74%	1.45%	2.11%
2	0.192	22.15	4.26	-0.33%	-1.59%	-1.81%	0.305	11.56	3.52	-0.08%	1.18%	0.99%
3	0.185	22.11	4.10	-0.81%	-0.71%	-1.40%	0.294	11.36	3.34	-0.44%	-2.49%	-3.01%
4	0.196	22.15	4.35	0.70%	-1.94%	-1.14%	0.306	11.34	3.46	-0.78%	-0.28%	-1.01%
5	0.193	22.20	4.29	0.04%	-1.37%	-1.23%	0.302	11.04	3.34	-0.90%	-3.43%	-4.40%
6	0.196	21.25	4.16	0.51%	-5.93%	-5.34%	0.305	11.04	3.36	-1.05%	-2.94%	-3.91%
7	0.194	23.08	4.48	0.55%	2.53%	3.21%	0.305	11.47	3.49	-0.09%	0.34%	0.14%
Feeder 2												
8	0.111	8.87	0.99	-0.46%	0.85%	0.84%	0.181	1.90	0.34	-0.72%	2.00%	0.83%
9	0.118	9.86	1.16	0.04%	-0.01%	0.34%	0.192	2.04	0.39	0.10%	1.17%	0.72%
10	0.119	10.29	1.22	-0.84%	0.87%	0.36%	0.195	2.06	0.40	0.13%	-0.39%	0.65%
Feeder 3												
11	0.189	23.29	4.40	0.06%	0.34%	0.32%	0.300	11.52	3.46	0.74%	-1.75%	-0.95%
12	0.188	22.13	4.16	0.64%	-4.34%	-3.84%	0.295	11.49	3.39	0.03%	-2.48%	-2.59%
13	0.187	22.50	4.20	-0.19%	-2.74%	-3.03%	0.296	10.96	3.24	0.27%	-6.94%	-6.82%
14	0.182	23.11	4.20	0.45%	0.85%	1.22%	0.288	11.67	3.36	1.03%	-2.92%	-2.04%
15	0.187	23.41	4.39	0.18%	1.20%	1.28%	0.295	12.27	3.62	0.03%	4.17%	4.05%
16	0.187	22.93	4.29	3.44%	0.10%	3.47%	0.295	11.24	3.32	3.55%	-6.51%	-3.32%
17	0.181	23.06	4.18	-3.04%	-0.28%	-3.42%	0.285	11.31	3.23	-3.36%	-3.97%	-7.33%
Feeder 4												
18	0.197	22.70	4.48	0.09%	2.68%	2.67%	0.309	11.30	3.50	-0.58%	0.66%	0.15%
19	0.189	21.62	4.09	-0.88%	-1.11%	-2.10%	0.303	11.42	3.46	0.58%	-0.18%	0.50%
20	0.200	22.50	4.50	1.44%	1.77%	3.13%	0.312	11.33	3.53	0.18%	0.85%	1.11%
21	0.197	21.75	4.28	-0.05%	-1.64%	-1.78%	0.312	11.40	3.56	0.46%	1.50%	2.04%
22	0.190	21.24	4.04	-0.53%	-2.84%	-3.46%	0.301	12.17	3.66	-0.17%	6.39%	6.32%
23	0.198	21.46	4.26	0.76%	-2.93%	-2.30%	0.310	11.18	3.46	-0.41%	-0.43%	-0.77%
24	0.195	21.60	4.22	-0.80%	-2.32%	-3.20%	0.311	10.94	3.40	0.04%	-2.62%	-2.51%
25	0.190	21.94	4.16	-0.63%	0.36%	-0.38%	0.300	11.30	3.39	-0.22%	-1.20%	-1.32%
Feeder 5												
26	0.117	10.49	1.22	0.66%	5.10%	5.58%	0.188	2.02	0.38	-0.33%	-1.07%	-2.52%
27	0.120	10.51	1.26	1.53%	1.79%	3.16%	0.193	2.09	0.40	0.34%	0.29%	0.47%
28	0.110	8.49	0.93	0.06%	-4.51%	-4.65%	0.178	1.93	0.34	-0.37%	3.28%	1.31%
Feeder 6												
29	0.118	8.49	1.01	0.38%	-0.05%	0.51%	0.192	1.84	0.35	0.23%	1.40%	0.91%
30	0.126	9.37	1.18	1.22%	-1.68%	-0.33%	0.204	2.15	0.44	0.83%	9.02%	9.37%
31	0.118	9.08	1.07	0.14%	6.89%	7.23%	0.194	1.80	0.35	1.10%	-0.43%	-0.05%
Feeder 7												
32	0.190	22.48	4.27	-0.47%	-2.28%	-2.67%	0.301	11.55	3.47	-0.43%	-0.21%	-0.81%
33	0.191	22.70	4.33	-0.16%	-1.32%	-1.42%	0.302	11.53	3.48	-0.12%	-0.36%	-0.65%
34	0.182	22.48	4.10	-0.37%	-0.91%	-1.22%	0.289	11.84	3.43	0.10%	-0.26%	-0.15%
35	0.192	23.29	4.46	0.30%	1.25%	1.62%	0.300	12.11	3.63	-0.62%	4.68%	3.86%
36	0.184	22.28	4.10	0.49%	-1.81%	-1.27%	0.290	12.36	3.59	0.49%	4.13%	4.65%
37	0.191	23.01	4.38	-0.26%	0.03%	-0.16%	0.300	11.06	3.32	-0.58%	-4.43%	-5.14%
38	0.185	22.16	4.09	0.84%	-2.33%	-1.46%	0.287	12.25	3.52	-0.54%	3.21%	2.67%

mance indices and individual feeder system indices are very close to the results obtained by an analytical method [6]. The results indicate that MCS results for the system indices are in good agreement with the analytical values. These studies clearly indicate that MCS is a powerful and accurate tool for performing distribution system reliability assessment.

Figs. 2–4 show the simulation results in terms of the probability distributions of system performance indices SAIFI, SAIDI and CAIDI for both cable and overhead line systems. Typical results for these system indices from the simulation studies are included in Table 3.

• The values of the system performance index SAIFI in each simulation were arranged in groups within an interval of 1.20 occ. customer-year^{-1} in which the index varies. This interval was divided into classes with width 0.06 occ. customer-year^{-1} up to 1.20 occ. customer-year^{-1}. It can be seen from Fig. 2 that the probability distributions of SAIFI for case A, C and E are almost same, so are the probability distributions of SAIFI for case B, D and F. Table 3 shows the cumulative probabilities of SAIFI for six cases for both line and cable systems. It can be seen from Table 3 that the probabilities of SAIFI greater than 0.24 occ. customer year^{-1} for cases A, C and E are 0.64694, 0.64956 and 0.64728 respectively; 0.9977, 0.99738 and 0.9981 for cases B, D and F respectively. These results indicate that not only the mean values of SAIFI for cases A, C

Table 2
System performance indices and individual feeder system indices

BUS 4	CABLES			LINES		
	SAIFI	SAIDI	CAIDI	SAIFI	SAIDI	CAIDI
a) base case A: disconnects--fuses--alternative supply--repair of transformer						
F1	0.191 (-0.04%)	4.25 (0.98%)	22.23 (0.88%)	0.301 (-0.42%)	3.44 (-0.96%)	11.43 (-0.63%)
F2	0.116 (0.71%)	1.13 (-0.49%)	9.69 (-0.60%)	0.189 (-0.33%)	0.38 (-0.16%)	2.00 (1.18%)
F3	0.187 (-0.35%)	4.27 (0.72%)	22.87 (1.04%)	0.295 (0.29%)	3.41 (-1.77%)	11.56 (-2.12%)
F4	0.195 (-0.14%)	4.28 (0.55%)	21.90 (0.59%)	0.308 (-0.08%)	3.52 (1.27%)	11.45 (1.34%)
F5	0.116 (-0.47%)	1.14 (-1.69%)	9.86 (-1.11%)	0.186 (-0.29%)	0.38 (1.49%)	2.01 (0.70%)
F6	0.121 (-0.59%)	1.08 (-2.26%)	8.98 (-1.50%)	0.197 (0.90%)	0.38 (2.75%)	1.93 (3.33%)
F7	0.188 (-0.10%)	4.27 (0.86%)	22.71 (0.85%)	0.297 (-0.32%)	3.49 (0.51%)	11.74 (0.61%)
SYST	0.190 (-0.16%)	4.26 (0.68%)	22.39 (0.83%)	0.300 (0.04%)	3.46 (-0.23%)	11.54 (-0.21%)
b) case B: no disconnects--no fuses--no alternative supply--repair of transformers						
F1	0.455 (0.09%)	31.02 (1.55%)	68.24 (1.45%)	0.675 (-0.05%)	23.40 (-1.88%)	34.69 (-1.82%)
F2	0.174 (0.17%)	5.17 (0.89%)	29.79 (0.72%)	0.283 (-0.01%)	1.43 (0.99%)	5.07 (1.38%)
F3	0.448 (-0.22%)	30.94 (1.04%)	69.06 (1.27%)	0.659 (-0.42%)	23.26 (-2.22%)	35.29 (-1.79%)
F4	0.497 (-0.51%)	33.67 (4.62%)	67.80 (5.17%)	0.739 (1.32%)	25.03 (-7.47%)	33.89 (-8.65%)
F5	0.169 (1.55%)	5.11 (1.02%)	30.16 (-0.53%)	0.278 (-0.60%)	1.39 (-0.90%)	4.98 (-0.30%)
F6	0.171 (0.55%)	5.18 (-0.29%)	30.25 (-0.83%)	0.278 (-0.77%)	1.48 (6.00%)	5.34 (6.80%)
F7	0.450 (-0.23%)	31.09 (0.74%)	69.08 (0.98%)	0.665 (0.01%)	23.07 (-3.08%)	34.68 (-3.10%)
SYST	0.463 (-0.22%)	31.69 (2.11%)	68.48 (2.37%)	0.685 (0.44%)	23.68 (-3.90%)	34.56 (-4.35%)
c) case C: no disconnects--fuses--no alternative supply--repair of transformers						
F1	0.190 (0.28%)	8.24 (0.55%)	43.24 (0.31%)	0.302 (0.10%)	4.45 (0.44%)	14.72 (0.13%)
F2	0.117 (0.15%)	3.51 (-0.37%)	30.07 (-0.23%)	0.190 (0.01%)	0.96 (1.41%)	5.07 (1.40%)
F3	0.186 (-0.16%)	8.08 (0.71%)	43.39 (0.70%)	0.293 (-0.22%)	4.22 (-3.85%)	14.39 (-3.82%)
F4	0.194 (0.51%)	8.45 (-0.61%)	43.56 (-1.10%)	0.308 (-0.03%)	4.40 (-1.26%)	14.31 (-1.41%)
F5	0.115 (0.37%)	3.38 (1.63%)	29.54 (1.56%)	0.188 (0.32%)	0.94 (0.70%)	4.99 (-0.15%)
F6	0.121 (-0.56%)	3.62 (-0.69%)	30.04 (-0.14%)	0.195 (-0.20%)	0.97 (-0.66%)	5.00 (0.04%)
F7	0.189 (-0.36%)	8.12 (0.96%)	43.05 (1.14%)	0.297 (-0.50%)	4.38 (-0.58%)	14.79 (-0.30%)
SYST	0.190 (0.07%)	8.22 (0.36%)	43.29 (0.20%)	0.300 (0.01%)	4.36 (-1.32%)	14.54 (-1.38%)
d) case D: disconnects--no fuses--alternative supply--repair of transformers						
F1	0.454 (0.28%)	6.40 (2.92%)	14.11 (2.68%)	0.674 (-0.16%)	5.03 (-1.72%)	7.47 (-1.49%)
F2	0.174 (-0.12%)	1.40 (-7.28%)	8.05 (-7.42%)	0.282 (-0.43%)	0.46 (-1.64%)	1.64 (-1.14%)
F3	0.449 (-0.42%)	6.65 (-0.84%)	14.80 (-0.50%)	0.666 (0.56%)	5.02 (-2.14%)	7.54 (-2.56%)
F4	0.496 (-0.32%)	7.50 (2.59%)	15.13 (2.87%)	0.740 (1.48%)	6.10 (0.99%)	8.25 (-0.41%)
F5	0.170 (1.06%)	1.29 (0.01%)	7.58 (-1.04%)	0.280 (-0.09%)	0.48 (1.84%)	1.71 (2.46%)
F6	0.171 (0.67%)	1.27 (-4.09%)	7.44 (-4.88%)	0.279 (-0.36%)	0.45 (0.00%)	1.59 (-1.24%)
F7	0.451 (-0.36%)	6.96 (-0.16%)	15.45 (0.14%)	0.663 (-0.28%)	5.32 (-2.03%)	8.02 (-1.80%)
SYST	0.463 (-0.14%)	6.90 (1.07%)	14.91 (1.37%)	0.686 (0.63%)	5.39 (-0.94%)	7.85 (-1.60%)
e) case E: disconnects--fuses--alternative supply--replacement of transformers						
F1	0.192 (-0.62%)	1.45 (-0.69%)	7.55 (-0.48%)	0.300 (-0.30%)	0.63 (1.29%)	2.09 (2.05%)
F2	0.116 (0.71%)	1.11 (0.99%)	9.55 (0.87%)	0.190 (-0.12%)	0.38 (0.47%)	2.01 (1.57%)
F3	0.187 (-0.68%)	1.48 (-1.69%)	7.88 (-0.99%)	0.293 (-0.44%)	0.66 (7.04%)	2.27 (7.44%)
F4	0.194 (0.32%)	1.46 (-0.52%)	7.50 (-1.05%)	0.307 (-0.20%)	0.67 (6.24%)	2.18 (7.27%)
F5	0.113 (1.49%)	1.13 (-0.56%)	9.94 (-1.91%)	0.186 (-0.30%)	0.37 (-0.00%)	1.99 (-0.75%)
F6	0.121 (-0.80%)	1.08 (-1.55%)	8.90 (-0.56%)	0.197 (0.90%)	0.37 (-1.08%)	1.86 (-1.01%)
F7	0.189 (-0.31%)	1.46 (-0.27%)	7.76 (-0.04%)	0.296 (-0.60%)	0.63 (1.73%)	2.13 (1.87%)
SYST	0.191 (-0.29%)	1.46 (-0.75%)	7.67 (-0.61%)	0.300 (-0.10%)	0.67 (7.58%)	2.23 (7.54%)
f) case F: disconnects--no fuses--no alternative supply--repair of transformers						
F1	0.458 (-0.66%)	15.67 (0.01%)	34.21 (0.67%)	0.676 (0.11%)	11.60 (1.01%)	17.16 (0.88%)
F2	0.173 (0.31%)	3.63 (0.58%)	20.92 (0.39%)	0.280 (-0.92%)	1.11 (6.56%)	3.95 (7.68%)
F3	0.448 (-0.17%)	15.25 (1.28%)	34.07 (1.48%)	0.660 (-0.33%)	11.21 (-1.62%)	16.98 (-1.26%)
F4	0.494 (0.00%)	17.95 (-0.21%)	36.33 (-0.23%)	0.739 (1.37%)	13.49 (0.42%)	18.25 (-0.92%)
F5	0.171 (0.64%)	3.65 (0.70%)	21.38 (0.18%)	0.280 (-0.00%)	1.07 (2.98%)	3.83 (2.55%)
F6	0.171 (0.41%)	3.58 (0.26%)	20.90 (-0.01%)	0.279 (-0.41%)	1.05 (2.45%)	3.75 (2.68%)
F7	0.449 (-0.08%)	17.65 (1.15%)	39.27 (1.22%)	0.665 (0.04%)	13.38 (0.90%)	20.11 (0.85%)
SYST	0.463 (-0.13%)	16.71 (0.56%)	36.11 (0.74%)	0.686 (0.56%)	12.48 (0.27%)	18.20 (-0.25%)

The numbers in parenthesis are the percentage differences with respect to the analytical values.

and E (cases B, D and F) are same but also their probability distributions are almost similar. At the same time, the system SAIFI for cases A, C and E are seen to be greatly reduced when compared with those of the other cases. This is due to the presence of fuses in the laterals. SAIFI are therefore not affected by the presence of the alternative supply, presence of disconnects and by the repair or replacement of the transformers. This is because SAIFI is only dependent on the component failure time distribution and on the number of customers served at each load point. Because the mean values of SAIFI for both groups of cases A, C, E and cases B, D, F for line systems are 0.300 and 0.685 occ. customer-year^{-1} respectively, we can see from Table 3 and Fig. 2 that SAIFIs are densely distributed around their mean values.

From Table 3, the probability of SAIFI greater than 0.24 occ. customer-year^{-1} for case A for both line and

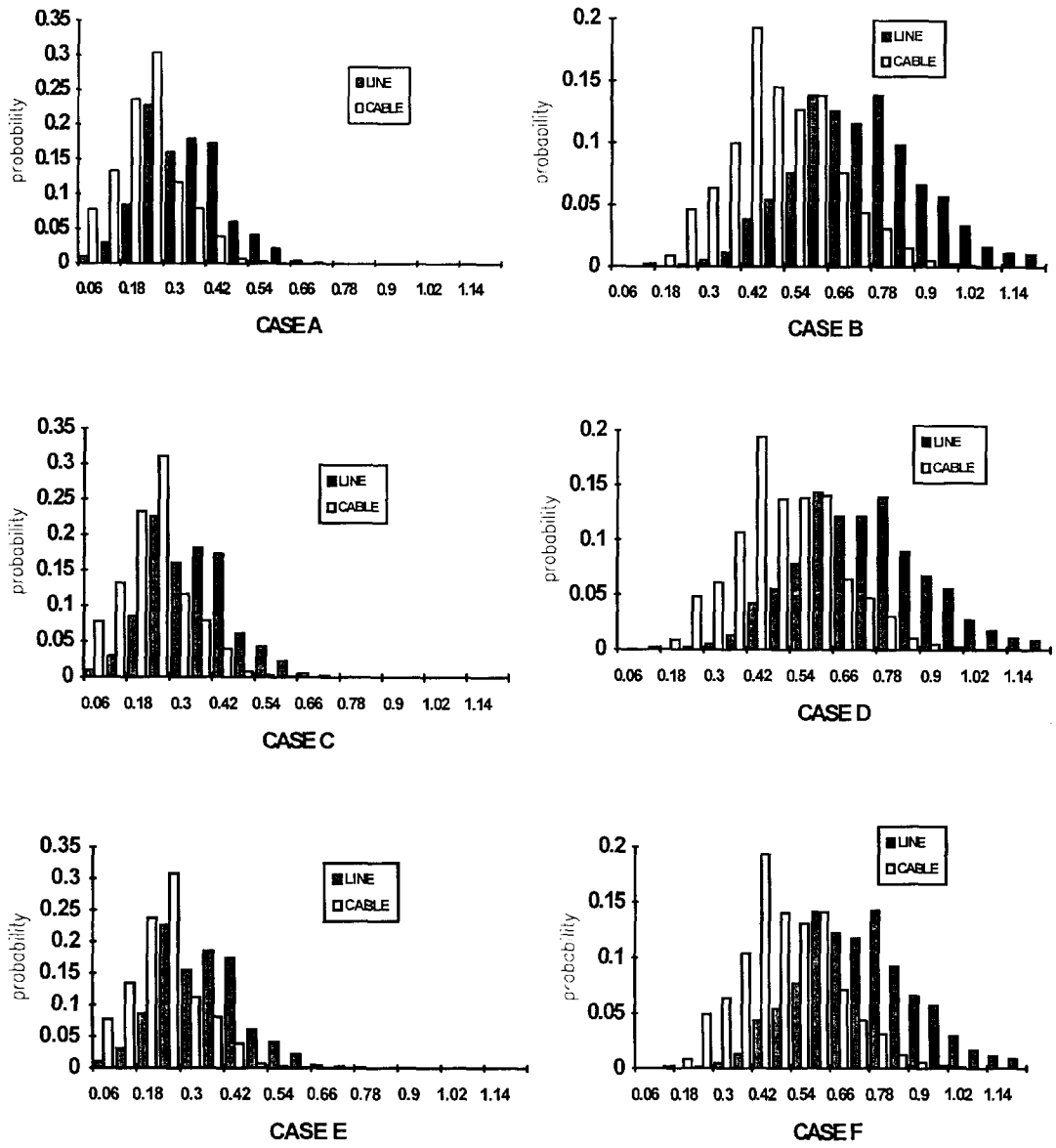


Fig. 2. Probability distributions of SAIFI for cable and overhead line distribution systems.

cable systems are 0.64694 and 0.24782 respectively. This means that the 'line system' failure rates are higher than those for the 'cable system'. Similar conclusions can be drawn from the other 5 cases. This is due to the higher failure rates of overhead lines.

• The values of the system performance index SAIDI in each simulation were arranged in groups within an interval of 60 hr customer-year^{-1} in which the index varies. This interval was divided into classes with width 3.00 h customer-year^{-1} up to 60.00 h customer-year^{-1}. Fig. 3 shows the probability distributions of SAIDI for six cases for line systems. Table 3 shows the cumulative probabilities of SAIDI for both line and cable systems. It can be seen from Table 3 that the decreasing order of probabilities of SAIDI greater than 12.0, 24.0, 36.0 and 48.0 h customer-year^{-1} for line systems is B, F, D, C, A and E, i.e. case B has the highest cumulative probability for a specified outage duration, while case E has the lowest. This result corresponds to the decreasing order of mean values of SAIDI for the six cases (see Table 2). It can be seen from Table 3 that the decreasing order of probabilities of SAIDI greater than 12.0 h customer-year^{-1} for cable systems is B, F, C, D, A and E which corresponds to the decreasing order of mean values of SAIDI for the six cases (see Table 2). The decreasing order of probabilities of SAIDI greater than 24.0, 36.0 and 48.0 h customer-year^{-1} for cable systems, however, is B, F, D, C, A and E. This means that under some conditions the actual system performance cannot be reflected by average expected values.

It can also be seen from Table 3 that the probabilities of SAIDI greater than 12.0, 24.0, 36.0 and 48.0 h

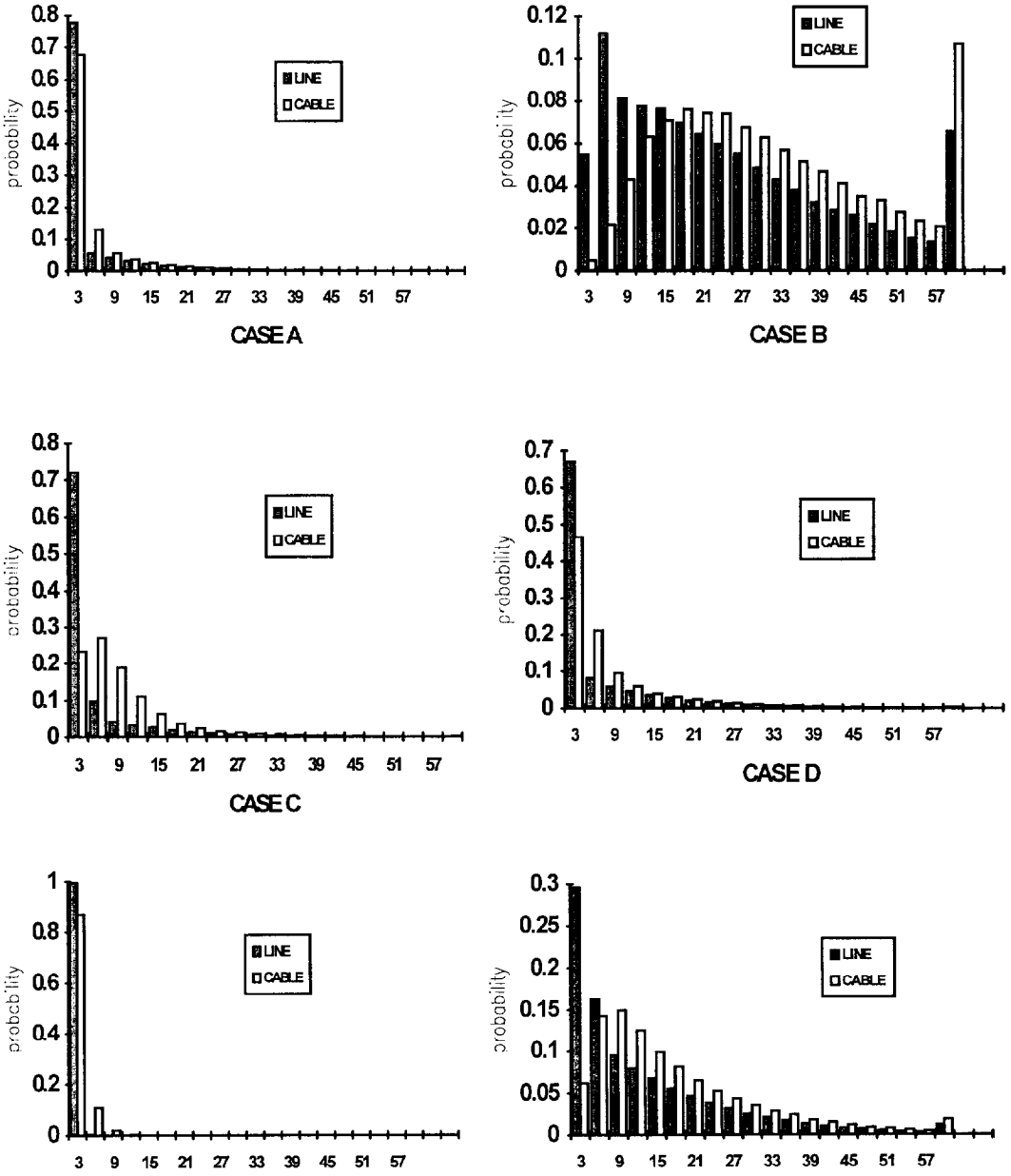


Fig. 3. Probability distributions of SAIDI for cable and overhead line distribution systems.

customer-year^{-1} for case F for both line and cable systems are higher than the corresponding values for case D, and case B higher than case F; hence the SAIDI for case F is higher than that for case D due to alternative supply not being available. SAIDI for case B is higher than that for case F which is due to disconnects not being present. The ability to backfeed has a pronounced effect on the length of the interruption. This effect, however, would be reduced significantly if the probability of backfeed being available is low.

The probability distributions of SAIDI are different for each case. This is because the SAIDI distribution is dependent on the repair time and restoration time distributions. Due to the large difference of repair time and restoration time for each case, the probability distributions of SAIDI for each case are very different, as shown in Fig. 3. Hence the distribution system SAIDI is significantly affected by any change in the operating philosophy, especially by the inclusion of fuses in laterals and by the replacement of failed transformers in the laterals.

• The values of the system performance index CAIDI in each simulation are arranged in groups within an interval of 120 h customer-year^{-1} in which the index varies. This interval was divided into classes with width 6.00 h customer-year^{-1} up to 120.00 h customer-year^{-1}. Fig. 4 shows the probability distributions of CAIDI

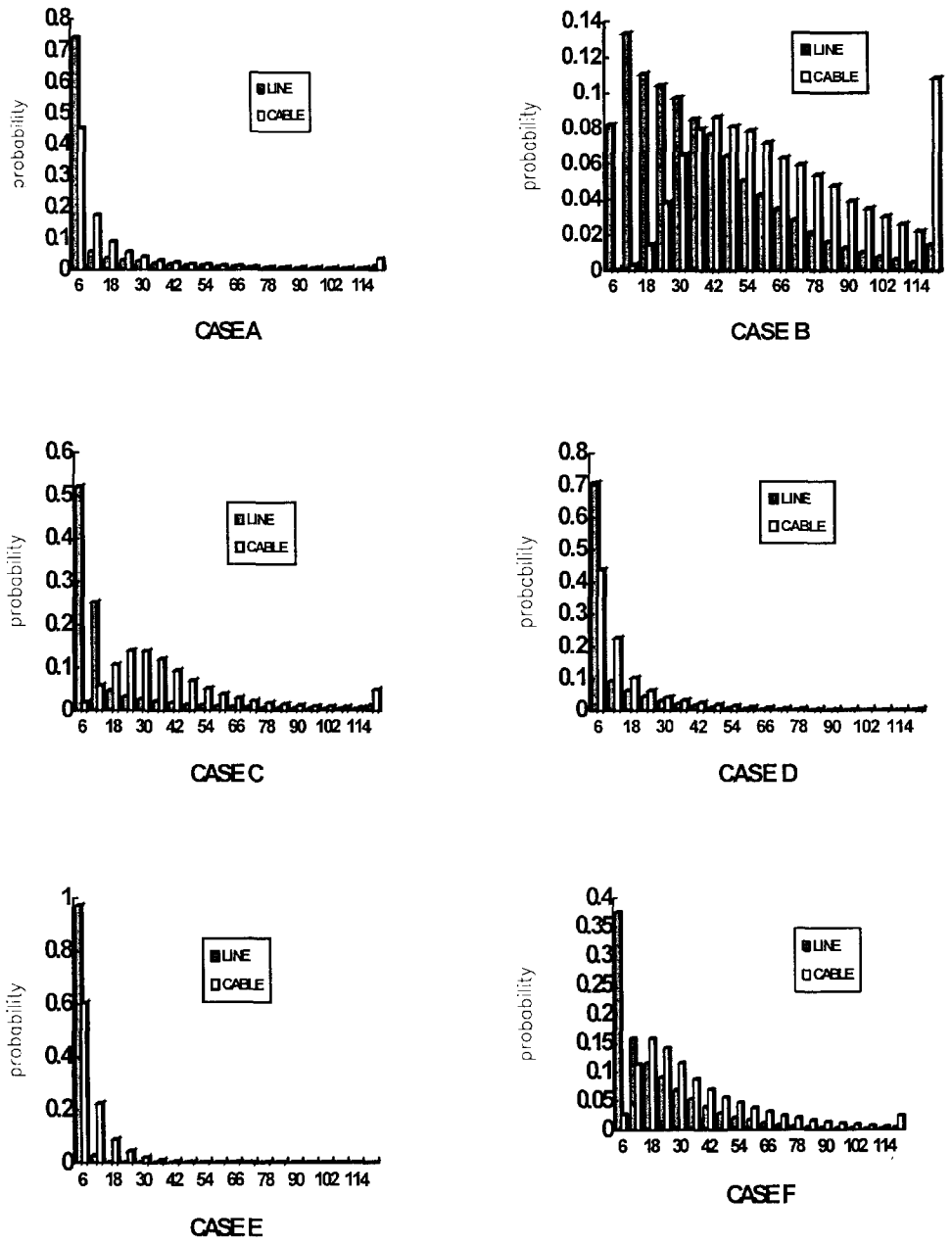


Fig. 4. Probability distributions of CAIDI for cable and overhead line distribution systems.

for all the six cases for overhead line systems. Fig. 4 is very similar with Fig. 3. This can be explained from the fact that CAIDI is a ratio of the SAIDI and the SAIFI, and the SAIFI for all the six cases are densely distributed around their mean values (see Fig. 2). If SAIFI is a constant, the shape of the probability distributions of CAIDI is the same as the shape of the probability distributions of SAIDI. For large systems the dispersion of the SAIFI distributions tends to decrease, hence the shape of the distributions of CAIDI is similar with the shape of the distributions of SAIDI.

Table 3 shows the cumulative probabilities of CAIDI for both line and cable distribution systems. From this table, the decreasing order of probabilities of CAIDI greater than 12.0, 24.0, 36.0, 48.0, 72.0 and 96.0 h cust.int^{-1} is B, F, C, A, D and E for overhead line systems, and B, C, F, A, D and E for cable systems. This means that the absence of fuses has a bigger effect on CAIDI than the absence of disconnects has for line systems. The cable case can be explained in a similar manner. It can be seen from Table 3 that the probabilities of CAIDI greater than 12.0 h cust.int^{-1} for cases A and D are very close, 0.20252 and 0.20058 respectively, i.e. the probabilities of CAIDI less than 12.0 h cust.int^{-1} for these two cases are 0.79748 and 0.79942 respectively. Hence under the conditions of CAIDI <

Table 3
Cumulative probabilities of SAIFI, SAIDI and CAIDI for both cable and line systems

	CASE A		CASE B		CASE C		CASE D		CASE E		CASE F	
	CABLES	LINES	CABLES	LINES	CABLES	LINES	CABLES	LINES	CABLES	LINES	CABLES	LINES
SAIFI (occ/cust.-yr)												
>0.24	0.24782	0.64694	0.94244	0.99770	0.24702	0.64956	0.93938	0.99738	0.24330	0.64726	0.93970	0.99810
>0.48	0.00450	0.07202	0.44130	0.88834	0.00378	0.07404	0.43972	0.88092	0.00386	0.07154	0.43880	0.88268
>0.72	0.00004	0.00094	0.05642	0.43222	0.00002	0.00092	0.04999	0.41644	0.00000	0.00100	0.05296	0.42458
>0.96	0.00002	0.00000	0.00188	0.07194	0.00002	0.00000	0.00122	0.06426	0.00000	0.00002	0.00138	0.06628
SAIDI (hr./cust.-yr)												
>12	0.10130	0.09070	0.86750	0.67492	0.19368	0.10264	0.16824	0.14550	0.00082	0.00020	0.52030	0.36676
>24	0.03094	0.02778	0.57260	0.40544	0.05096	0.03160	0.05634	0.04898	0.00022	0.00018	0.22282	0.15746
>36	0.00988	0.00094	0.33458	0.22138	0.01614	0.00932	0.01862	0.01616	0.00022	0.00016	0.09142	0.06300
>48	0.00286	0.00000	0.17874	0.11228	0.00480	0.00286	0.00620	0.00564	0.00022	0.00016	0.03776	0.02424
CAIDI (hr./cust.int)												
>12	0.37616	0.20252	0.99702	0.78506	0.92292	0.22910	0.33858	0.20058	0.17399	0.00240	0.86216	0.46776
>24	0.23228	0.13814	0.94480	0.57054	0.67810	0.15164	0.17580	0.09550	0.04530	0.00038	0.56404	0.26406
>36	0.16365	0.09544	0.80072	0.38860	0.42117	0.10470	0.10454	0.04562	0.01475	0.00028	0.36260	0.14424
>48	0.12434	0.06684	0.63364	0.24792	0.25934	0.07344	0.06288	0.02250	0.00626	0.00022	0.23786	0.07678
>72	0.07542	0.03330	0.36130	0.09216	0.12159	0.03654	0.02414	0.00602	0.00140	0.00018	0.10056	0.02084
>96	0.04840	0.01718	0.18650	0.03238	0.06959	0.01906	0.00990	0.00192	0.00042	0.00016	0.04226	0.00652

12.0 h cust.int^{-1}, the system performance for these two cases is similar. But for the condition of CAIDI > 24.0 h cust.int^{-1}, the system reliability for case D is not as good as that for case A. This kind of performance assessment of the distribution systems cannot be reflected well by analytical indices. The MCS method is therefore a very useful tool in this regard.

4. Conclusions

This paper illustrates the application of a Monte Carlo simulation technique to a comprehensive test distribution system. Some changes in operating philosophy and configuration are modeled in the paper. Basic load point indices, system performance indices, individual feeder indices and their distributions by using the MCS method are presented and compared with those obtained using an analytical approach. The relative errors between the simulation and analytical methods are shown to be very small under certain specified conditions. The Monte Carlo simulation method is therefore a useful and powerful tool for evaluating the reliability indices not just their average values but also their probability distributions. The information obtained from these studies can be used by the distribution system engineer/planner for system reinforcements and for any cost/benefit assessment.

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